Part D, D1 (a),(b),(c): independent rederivation (auditor 1)

2026-10-28

Inputs: Def c1-pairing, Def c1-bracket, Prop c1-Q, Def c3-delta, Def c3-W, Prop c3-QP, Def c7-vdot, Lemma c7-weights, Prop c7-dil; [C07, §6.2–6.4]. Checks: `bv_toy.py` (exact, finite-dimensional, $\mathbb{Z}/2$-graded, odd pairing, Koszul signs; 79 checks, 0 failures), `n4_scaling_check.py` (sympy, $\mathbb{R}^4$ kernels; 9 checks, 0 failures). $S_M := S^{(1)}(\Delta)$, $M := v$. for $v = 2E$, D_M the derivation induced by M [C07, §6.2], $W'_J(X) := e^{-W(P,J)/\hbar} e^{\hbar\partial_P}(Xe^{J/\hbar})$, $P = P(\varepsilon,\infty)$, $K_\varepsilon = (Q\otimes 1 + 1\otimes Q)P$.

0. Sign of the bracket with S_M

$\{\frac{1}{2}\langle e, Qe\rangle, -\} = D_Q$ with Q of Prop c1-Q $\iff$ $\Delta_0 = -\partial_{K_0} = \sum_{\text{pairs}} \dfrac{\partial}{\partial(\text{even member})}\dfrac{\partial}{\partial(\text{odd member})}$ (left derivatives)

$$S_M = \int \mathrm{dvol}\Big((MA, A^*) - (M\lambda, \lambda^*) - (Mc, c^*)\Big) \implies \{S_M, F\} = +D_M F, \qquad \{S_M, F\}_\varepsilon = D_{e^{-\varepsilon H}M} F.$$

1. D1(a)

$$e^{\hbar\partial_P} S_M e^{-\hbar\partial_P} = S_M + \hbar\,(\partial_P S_M) + \hbar\, D_{P\star M} + \tfrac{\hbar^2}{2}\,\partial_{(P\star M\otimes 1 + 1\otimes P\star M)P}, \qquad \big[\partial_P, \partial_{(P\star M\otimes 1+1\otimes P\star M)P}\big] = 0,$$

$\partial_P S_M = 0$ (P even, S_M odd and quadratic: every monomial of S_M pairs opposite parities),

$$\big((P\star M\otimes 1 + 1\otimes P\star M)P\big)\star = 2\,P\star\circ M\circ P\star\,.$$

With κ the kernel of $P\star\circ M\circ P\star$ (graded symmetric, odd), $W = W(P,J)$, ξ odd, left derivative:

$$\frac{\partial}{\partial\xi} W(P, J+\xi S_M) = W'_J(S_M) = S_M + D_{P\star M}W + \hbar^2 e^{-W/\hbar}\partial_\kappa e^{W/\hbar} = S_M + D_{P\star M}W + \hbar\,\partial_\kappa W + \tfrac{1}{2}\{W,W\}_\kappa,$$

$$\{F,G\}_\kappa := \partial_\kappa(FG) - (\partial_\kappa F)G - (-1)^{|F|}F\,\partial_\kappa G.$$

Right ξ-derivative: $-W'_J(S_M)$. N=4 slots: $P\star\circ M\circ P\star$ maps $A^*|_V \to c$ and $c^* \to A|_V$ only, so $\kappa \in c\otimes A|_V \oplus A|_V\otimes c$; with $[M, P\star] = 4\varepsilon L_\varepsilon\star = 4\varepsilon\, Q^{GF}e^{-\varepsilon H}$,

$$P\star\circ M\circ P\star = (Q^{GF})^2\Big(\int_\varepsilon^\infty e^{-tH}dt\Big)^2 M + 4\varepsilon\,(Q^{GF})^2\Big(\int_\varepsilon^\infty e^{-tH}dt\Big)e^{-\varepsilon H}, \qquad (Q^{GF})^2A^* = 2d^*(A^*|_V),\ (Q^{GF})^2c^* = -2dc$$

2. D1(b)

$[Q, M] = 0 \Rightarrow QS_M = 0$; $QP\star + P\star Q = K_\varepsilon\star = e^{-\varepsilon H}$;

$$[Q, P\star M] = e^{-\varepsilon H}M, \qquad (Q\otimes 1 + 1\otimes Q)\kappa = \tfrac{1}{2}\big(e^{-\varepsilon H}M\otimes 1 + 1\otimes e^{-\varepsilon H}M\big)P - \tfrac{1}{2}\big(P\star M\otimes 1 + 1\otimes P\star M\big)K_\varepsilon.$$

$O_\varepsilon(J) := QJ + \frac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J$, $QW = W'_J(O_\varepsilon(J))$.

$$O_\varepsilon(J + \xi S_M) = O_\varepsilon(J) - \xi\big(\{S_M, J\}_\varepsilon + \hbar\Delta_\varepsilon S_M\big),$$

$$Q\,W'_J(S_M) = W'_J\big(\{S_M, J\}_\varepsilon\big) + \hbar\Delta_\varepsilon S_M - \hbar^{-1}\Big(e^{-W/\hbar}e^{\hbar\partial_P}\big(S_M\,O_\varepsilon(J)\,e^{J/\hbar}\big) - W'_J(S_M)\,W'_J(O_\varepsilon(J))\Big),$$

the last bracket connected (no negative power of $\hbar$), nonzero unless $O_\varepsilon(J) = 0$; $\hbar\Delta_\varepsilon S_M$ field independent. Term by term from §1:

$$Q\,W'_J(S_M) = D_{e^{-\varepsilon H}M}W - D_{P\star M}(QW) + \hbar^2\Big(e^{-W/\hbar}\partial_{(Q\otimes 1+1\otimes Q)\kappa}e^{W/\hbar} - \hbar^{-1}(QW)\,e^{-W/\hbar}\partial_\kappa e^{W/\hbar} - \hbar^{-1}e^{-W/\hbar}\partial_\kappa\big((QW)e^{W/}$$

3. D1(c)

$$[D_M, \partial_\phi] = -\partial_{(M\otimes 1+1\otimes M)\phi} \quad (\phi \text{ even or odd}), \qquad \big((M\otimes 1 + 1\otimes M)\phi\big)\star = [M, \phi\star],$$

$$(M\otimes 1 + 1\otimes M)P(\varepsilon,\infty) = 4\varepsilon L_\varepsilon = -4\varepsilon\frac{d}{d\varepsilon}P(\varepsilon,\infty) \quad (A\otimes A:\ 2+2,\ c\otimes A^*|_V:\ 0+6,\ \lambda\otimes\lambda:\ 3+3),$$

$$(M\otimes 1 + 1\otimes M)K_t = -4t\frac{d}{dt}K_t, \qquad E\big|_{\text{Fourier}} = -(4 + p\cdot\partial_p),$$

$$D_M W(P,J) = W'_J(D_M J) - \hbar^2 e^{-W/\hbar}\partial_{(M\otimes 1+1\otimes M)P}e^{W/\hbar} \implies \{S_M, W(P,J)\} = W'_J(\{S_M, J\}) + 4\varepsilon\frac{\partial}{\partial\varepsilon}\Big|_J W(P(\varepsilon,\infty), J).$$

Sign $+4\varepsilon\partial_\varepsilon$ confirmed; $-4\varepsilon\partial_\varepsilon$ fails (control). The bracket on the left is the scale-0 bracket; $\{S_M, W\}_\varepsilon = D_{e^{-\varepsilon H}M}W \neq D_M W$.